Structure-informed robust weighting for photometric visual servoing under occlusion: a pre-registered simulation study

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Abstract

Direct visual servoing feeds every pixel into the control law, so a partial occlusion biases the commanded velocity unless the corrupted measurements are down-weighted. We study a structure-informed robust weighting for Hermite photometric visual servoing, which normalises each residual by a local Gauss–Hermite gradient magnitude of the current image before applying Tukey's biweight. We show that, when the interaction matrix is computed from the current image, measurements in untextured regions have zero interaction rows, so their weights cannot affect the control. We also give a sufficient condition for local convergence under arbitrary time-varying weights. A pre-registered simulation study of 1 792 runs (32 initial poses, four occluder appearances, occluded fractions of 7.5–45 %, two textures) compares the weighting with Tukey weighting, a structure-penalising weighting and no weighting. With the desired-image interaction matrix and a budget of 600 iterations, the structure-informed weighting reduces the camera-centre error of Tukey weighting by a third (median ratio 0.66, 95 % CI 0.60–0.73, Holm-adjusted $p = 4.2 \times 10^{-7}$). The advantage is partly one of speed and shrinks as the budget grows. Robustness to large occluders depends on how the weighting is paired with the interaction matrix. With the desired-image matrix, the structure-informed weighting converges from 27–30 of 32 poses up to 45 % occlusion while Tukey weighting fails from every pose; with the current-image matrix the ordering reverses. On a second texture, Tukey weighting converges from every pose to a biased equilibrium 128–175 mm from

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the target, which the structure-informed weighting avoids.

Keywords: Visual servoing, photometric visual servoing, robust estimation, M-estimators, occlusion, Gauss–Hermite filters, pre-registration

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1. Introduction

Photometric, or direct, visual servoing uses the image intensities themselves as visual features [1, 2]. It dispenses with feature extraction, matching and tracking, and it reaches a positioning accuracy that geometric features rarely match, because every pixel contributes to the control law. The same property is its weakness: a region of the image that does not obey the model of the scene—an occluder, a specular highlight, a moving object—contributes as much as any other region, and it biases the commanded velocity in proportion to its size and contrast. Alternatives to raw intensities, such as mutual information [3], photometric Gaussian mixtures [4], photometric moments [5] and multi-scale Gauss–Hermite filter responses [6], change the representation but not this property.

The established remedy is robust estimation: each measurement receives a weight computed by an M-estimator from the magnitude of its residual, and the weights enter the control law [7, 8]. Such a weighting is *structure-blind*: it judges a measurement by its residual alone. Early in the servo, when the pose error is large, the residuals are large wherever the image is textured, and a residual-based estimator rejects valid, informative measurements together with the occluded ones. Direct visual odometry addresses the same heteroscedasticity by weighting residuals with a function of the image gradient [9, 10]. In DSO the weight decreases with the gradient magnitude [10]. The structure-informed weighting studied here does the opposite: it normalises each residual by a local gradient magnitude computed with the first-order Gauss–Hermite kernels of the servoing scheme, so that a large residual at a strongly textured position is treated as expected and a residual at an untextured position as suspect. In the terminology of robust regression it is a Schweppe-type generalised M-estimator [11, 12] whose design function reverses the usual bounded-influence direction.

Three questions are open. The first is *where* a weighting can act. The interaction matrix of a photometric feature is proportional to the image gradient. When it is computed from the current image, as in many implementations, measurements in a region that is uniform in the current image have

zero interaction rows, and no weight assigned to them can change the control. When it is computed from the desired image, the same measurements have non-zero rows. The choice between the two matrices is usually discussed in terms of the convergence domain [13, 14]; its consequence for robust weighting has not, to our knowledge, been analysed. The second question is whether structure-informed weighting improves the final accuracy, the convergence or both, and by how much. Evidence for robust photometric servoing commonly rests on a few runs from one or two initial poses, which cannot separate an effect from the variation between poses. The third is how the competing weightings break down as the occluded fraction of the image grows.

This paper makes three contributions.

1. An analysis of the weighted Levenberg–Marquardt photometric control law (Section 4): weights on measurements that are uniform in the current image are inert when the interaction matrix is taken from the current image (Proposition 1); local convergence holds for any time-varying weight sequence under a rank condition that can be checked online (Proposition 2); and the error-proportional gain of the scheme yields algebraic rather than geometric convergence on an unoccluded scene (Proposition 3).
2. A pre-registered simulation study of 1 792 runs (Section 6). The hypotheses, the primary tests, the multiple-comparison correction and the decision rule were committed to a version-controlled repository before the data were collected. The study comprises a confirmatory campaign (four weightings, two interaction matrices, four occluder appearances, 32 initial poses), a replication on a second texture and a sweep of the occluded fraction.
3. An account of when each weighting fails (Section 7). Tukey weighting retains occluded measurements whose residual happens to be small, which biases the equilibrium or causes divergence once the interaction matrix gives those measurements non-zero rows. The structure-informed weighting rejects an occluder by the absence of structure rather than by the size of its residual, but with the current-image matrix it fails late, along the weakly observable direction of a planar target. Its accuracy advantage at a fixed iteration budget shrinks as the budget grows, which the pre-registered outcome did not anticipate.

Section 2 reviews related work, Section 3 states the servoing scheme, Section 5 defines the weightings compared, Section 8 discusses the findings and their

limits, and Section 9 concludes.

2. Related work

Direct visual servoing. Photometric visual servoing [1, 2] regulates the difference between the current and desired images using the interaction matrix of the luminance, derived from the brightness-constancy assumption, and a Levenberg–Marquardt control law [15]. Subsequent work changed the feature to enlarge the convergence domain or to gain robustness: mutual information [3], which tolerates occlusions and illumination changes through its statistical formulation; photometric Gaussian mixtures [4]; and photometric moments [5]. The scheme studied here, Hermite photometric visual servoing (HER-PVS) [6], uses as features the responses of summed Gauss–Hermite kernels [16] at four scales. Its interaction matrix is obtained by filtering the luminance interaction matrix with the same kernels.

Robust weighting in visual servoing. Comport et al. [7] introduced M-estimation into image-based visual servoing: a diagonal weight matrix computed by iteratively reweighted least squares multiplies both the error and the interaction matrix, and the stability of the resulting law is discussed for weights that vary slowly. Malis and Marchand [8] compared robust estimators for real-time robot vision. In these works the weights depend on the residual alone. The interaction matrix used by the controller—current, desired or their mean [13, 14]—is chosen for convergence, and its interplay with the weighting is not examined.

Robust direct alignment. Direct image alignment [17] and direct visual odometry face the same problem. LSD-SLAM [9] normalises each photometric residual by a propagated variance that includes a gradient-dependent term. DSO uses a Huber norm together with a gradient-dependent weight $c^2/(c^2+\|\nabla I\|^2)$, which down-weights high-gradient pixels [10]. Both treat the residual as heteroscedastic. The structure-penalising weighting V4 of Section 5 transposes DSO's design weight to the servoing problem as a control condition.

Generalised M-estimation. Dividing a residual by a function of the corresponding design row before the influence function is applied defines a Schweppe-type generalised M-estimator [11, 12, 18]. Classical bounded-influence estimators of this type reduce the influence of high-leverage rows. The structure-informed weighting reverses that direction. The reversal is justified by a heteroscedastic model of the residual (Section 5), not by influence considerations.

Evaluation practice. Pre-registration—fixing hypotheses and analyses before observing the data—separates confirmatory from exploratory findings [19]. It is rare in robotics, where method comparisons are often reported from a handful of runs without a correction for multiple comparisons. The present study pre-registers its primary family and decision rule. It reports effect sizes with exact confidence intervals and labels every unplanned analysis as exploratory.

3. Hermite photometric visual servoing with weighting

3.1. Features

Let I denote the current grey-level image and I^* the desired one, both of 320×240 pixels. Write $d_n^\sigma(x) = (-1)^n \, (2^n n! \sqrt{\pi}\, \sigma)^{-1/2} H_n(x/\sigma)\, e^{-x^2/(2\sigma^2)}$ for the Hermite function of order n at scale σ, with H_n the Hermite polynomial, and $D_{n,m}^\sigma(x,y) = d_n^\sigma(x)\, d_m^\sigma(y)$ for the separable kernel of orders (n,m). The summed kernel of scale σ_k is

$$D_k = \sum_{n=0}^{4} \sum_{m=0}^{4} D_{n,m}^{\sigma_k}, \qquad (\sigma_1, \sigma_2, \sigma_3, \sigma_4) = (0.5,\ 1.1,\ 1.8,\ 2.8)\ \text{px}, \tag{1}$$

sampled on a 9×9 grid. The feature vector stacks the four filtered images at the $N = 300 \times 220 = 66\,000$ positions p of the image interior,

$$\mathbf{s}(I) = \Big[(D_k \star I)(p) \Big]_{k=1..4,\ p \in \Omega_0} \in \mathbb{R}^P, \qquad P = 4N, \tag{2}$$

where $\star$ denotes correlation and Ω_0 excludes a ten-pixel border. The error is $\mathbf{e} = \mathbf{s}(I) - \mathbf{s}(I^*)$, with entries e_j, $j = 1, \ldots, P$; the measurement j belongs to the scale $k(j)$ and the position p_j.

3.2. Interaction matrix

Under brightness constancy the luminance at an image point of normalised coordinates (x, y) has the interaction row $\boldsymbol{\ell}_I(p) = -\nabla I(p)^\top \mathbf{L}_\mathbf{x}(x, y, Z)$, where $\mathbf{L}_\mathbf{x} \in \mathbb{R}^{2\times 6}$ is the interaction matrix of an image point at depth Z [13, 2]. The target is planar and, at the desired pose, fronto-parallel at $Z^* = 1.3\,\text{m}$; following [6], $Z = Z^*$ is used throughout. Since correlation is linear and Z is constant, the interaction row of the feature $(D_k \star I)(p)$ is

$$\boldsymbol{\ell}_j = \Big(D_{k(j)} \star \boldsymbol{\ell}_I \Big)(p_j), \tag{3}$$

each of the six components being filtered separately. Stacking the rows gives the interaction matrix $\mathbf{L}(I) \in \mathbb{R}^{P\times 6}$. The controller uses one of two matrices,

$$\widehat{\mathbf{L}} \in \Big\{\mathbf{L}(I),\ \mathbf{L}(I^*)\Big\} : \tag{4}$$

the *current-image* matrix $\mathbf{L}(I)$, recomputed at every iteration, or the *desired-image* matrix $\mathbf{L}(I^*)$, computed once.

3.3. Weighted control law

Let $\mathbf{W} = \mathrm{diag}(w_1, \ldots, w_P)$ with $w_j \in [0, 1]$. The camera velocity screw is

$$\mathbf{v} = -\lambda\, \mathbf{K}_{\mathbf{W}}^{-1} \widehat{\mathbf{L}}^{\top} \mathbf{W}^2 \mathbf{e}, \qquad \begin{aligned} \mathbf{K}_{\mathbf{W}} &= \mathbf{H}_{\mathbf{W}} + \mu\ \mathrm{diag}(\mathbf{H}_{\mathbf{W}}) + \varepsilon\, \mathbf{I}_6, \\ \mathbf{H}_{\mathbf{W}} &= \widehat{\mathbf{L}}^{\top} \mathbf{W}^2 \widehat{\mathbf{L}}, \end{aligned} \tag{5}$$

with the floor $\varepsilon = \max\Big(10^{-6}\,\mathrm{tr}(\mathbf{H}_{\mathbf{W}})/6,\ 10^{-12}\Big)$. The weights multiply both the rows of $\widehat{\mathbf{L}}$ and the residuals, so the normal equations carry w_j^2 [7]. Setting $\mathbf{W} = \mathbf{I}$ recovers the unweighted law of [6] up to the floor. The gain and the damping follow the error-proportional law of [6],

$$\lambda = N_e^{0.2}\, \bar{e}, \qquad \mu = N_e^{-0.2}\, \bar{e}^{\,5}, \qquad \bar{e} = \|\mathbf{e}_I\|^2 / N_e, \tag{6}$$

where $\mathbf{e}_I = I^* - I$ is the intensity error over the whole image and N_e its squared norm at the first iteration. The velocity is applied for the sampling period $T = 40\,\mathrm{ms}$.

4. Analysis

This section states three properties of the law (5). Proofs are in Appendix A. Let $\mathbf{r}$ denote the camera pose, $\mathbf{r}^*$ the desired pose and $\delta\mathbf{r}$ the pose error in a local parametrisation of $SE(3)$ at $\mathbf{r}^*$, so that $\delta\mathbf{r}_{k+1} = \delta\mathbf{r}_k + T\mathbf{v}_k + O(T^2\|\mathbf{v}_k\|^2)$.

4.1. Where a weight can act

Proposition 1 (Inertness of weights on uniform regions)**.** *Let j be a measurement whose support—that of $D_{k(j)}$ combined with the gradient operator—lies in a region where the current image I is constant. If $\widehat{\mathbf{L}} = \mathbf{L}(I)$, then $\boldsymbol{\ell}_j = \mathbf{0}$, and the velocity (5) depends on w_j and e_j only through the robust location and scale that enter the other weights. If $\widehat{\mathbf{L}} = \mathbf{L}(I^*)$, then $\boldsymbol{\ell}_j = (D_{k(j)} \star \boldsymbol{\ell}_{I^*})(p_j)$, which is non-zero wherever the desired image has structure within the support.*

The consequence for occlusion is direct. With the current-image matrix, the interior of an untextured occluder exerts no force on the camera whatever its weights: the occluder acts only through a band around its boundary, of the width of the filter support, where the current image contains the occluder's own contour. Any weighting can then act on the occluder only in that band. With the desired-image matrix, the interior of the occluder carries the rows of the hidden texture together with the residuals $I - I^*$ it produces. These measurements pull the camera unless the weighting rejects them.

The norm of a luminance row is, to within a known factor, the image-gradient magnitude. With $Z = 1.3\,\mathrm{m}$, a focal length of $870\,\mathrm{px}$ and the principal point at the image centre, the ratio of the singular values of $\mathbf{L}_{\mathbf{x}}$ over Ω_0 does not exceed 1.023, so $\|\boldsymbol{\ell}_I(p)\|$ is proportional to $\|\nabla I(p)\|$ within $2.3\,\%$. A gradient magnitude computed at the scales of the features (Section 5) is therefore a proxy for the row norm of a measurement, that is, for how strongly it constrains the pose.

4.2. Convergence under reweighting

The weights of an iteratively reweighted scheme change at every iteration, so the stability arguments for a fixed weighting do not apply directly. Near the desired pose a statement that holds for any weight sequence is available. Define

$$\delta_{\mathbf{W}} = \frac{\mu \max_i (\mathbf{H}_{\mathbf{W}})_{ii} + \varepsilon}{\lambda_{\min}(\mathbf{H}_{\mathbf{W}})}. \tag{7}$$

Proposition 2 (Local convergence under arbitrary reweighting)**.** *Consider an unoccluded scene, so that* $\mathbf{e}(\mathbf{r}^*) = \mathbf{0}$*, and either choice* (4) *of* $\widehat{\mathbf{L}}$*. Suppose that along the trajectory* $\delta_{\mathbf{W}_k} \le \bar{\delta} < 1$*,* $T\lambda_k \le 1$*, and* $\lambda_{\min}(\mathbf{H}_{\mathbf{W}})/\operatorname{tr}(\mathbf{H}_{\mathbf{W}})$ *is bounded away from zero. Then*

$$\|\delta\mathbf{r}_{k+1}\| \le \big(1 - T\lambda_k(1-\bar{\delta})\big)\|\delta\mathbf{r}_k\| + O\big(\|\delta\mathbf{r}_k\|^2\big) \tag{8}$$

for every sequence of weights $\mathbf{W}_k$*, so the pose error decreases monotonically in a neighbourhood of* $\mathbf{r}^*$*.*

The weights enter (8) only through the rank condition. With μ small, as it is near convergence, $\delta_{\mathbf{W}}$ is governed by the floor ε, and the condition reduces to $\lambda_{\min}(\mathbf{H}_{\mathbf{W}})/(\operatorname{tr}(\mathbf{H}_{\mathbf{W}})/6) > 10^{-6}$. This *rank margin* is logged at every iteration of the study. Under occlusion the same argument applies about the equilibrium $\bar{\mathbf{r}}$ of the weighted law, provided the matrix used by the

controller approximates the Jacobian of the error there; the final pose error then measures the bias $\|\bar{\mathbf{r}} - \mathbf{r}^*\|$ of the weighted estimator, not a truncated convergence.

4.3. Rate under the error-proportional gain

Proposition 3 (Algebraic convergence on an unoccluded scene)**.** *Under the assumptions of Proposition 2, let in addition* $\delta_{\mathbf{W}} \to 0$ *and* $\|\mathbf{e}_I\|^2 = \delta\mathbf{r}^\top \mathbf{Q}\, \delta\mathbf{r} + o(\|\delta\mathbf{r}\|^2)$ *with* $\mathbf{Q} \succ 0$*. Then, with* $c = TN_e^{-0.8}$ *and* $\rho_k = \delta\mathbf{r}_k^\top \mathbf{Q}\, \delta\mathbf{r}_k$*,*

$$\frac{1}{\rho_{k+1}} = \frac{1}{\rho_k} + 2c + O(\rho_k), \qquad \textit{hence} \qquad \rho_k \sim \frac{1}{2ck}, \qquad \|\delta\mathbf{r}_k\| = \Theta\Big(k^{-1/2}\Big). \tag{9}$$

If instead the scene is occluded, the weights are held fixed and $\|\mathbf{e}_I\|^2 \to e_\infty > 0$*, then* $\lambda \to \lambda_\infty = e_\infty N_e^{-0.8} > 0$ *and the convergence to the equilibrium of the weighted law is geometric, with factor at most* $1 - T\lambda_\infty(1 - \bar{\delta})$ *per iteration.*

The gain (6) vanishes with the error, so on a clean scene the servo slows down as it converges, while a persistent occlusion residual keeps the gain finite. Proposition 3 analyses this particular law. It is not a statement about photometric servoing in general, and it predicts that the final error after a fixed number of iterations is larger on a clean scene than on an occluded one.

5. Weightings compared

All weightings use Tukey's biweight [20] $w^{\mathrm{T}}(u) = (1 - u^2/c^2)^2$ for $|u| \le c$ and 0 otherwise, with $c = 4.6851$, applied to a median-centred, MAD-scaled argument [21],

$$u_j(\mathbf{x}) = \frac{x_j - \operatorname{med}_i x_i}{\hat{s}(\mathbf{x})}, \qquad \hat{s}(\mathbf{x}) = 1.4826\, \operatorname{med}_j \Big| x_j - \operatorname{med}_i x_i \Big|, \tag{10}$$

as implemented in ViSP [22]. Because w_j multiplies both the residual and the row in (5), the implemented estimator has the influence function $u(1-u^2/c^2)^4$ on $|u| \le c$; its efficiency at the Gaussian is 0.87 rather than the biweight's 0.95. The structure-informed weightings use the first-order Gauss–Hermite responses of the current image at the three coarser scales,

$$G(p) = \Big(\sum_{k=2}^{4} \Big(D_{1,0}^{\sigma_k} \star I \Big)(p)^2 + \Big(D_{0,1}^{\sigma_k} \star I \Big)(p)^2 \Big)^{1/2}, \qquad \bar{G}(p) = \frac{G(p)}{\frac{1}{N} \sum_{q \in \Omega_0} G(q)}. \tag{11}$$

Table 1: Weightings compared. $\bar{G}_j = \bar{G}(p_j)$; all four channels at a position share the structure factor.

	Name	Weight w_j	Role
V1	none	1	unweighted baseline
V2	Tukey	$w^{\mathrm{T}}(u_j(\mathbf{e}))$	residual-based reference [7]
V3	structure-informed	$w^{\mathrm{T}}(u_j(\tilde{\mathbf{e}}))$, $\tilde{e}_j = e_j / \max(\bar{G}_j, 0.1)$	method under study
V4	structure-penalised	$w^{\mathrm{T}}(u_j(\mathbf{e}))\,(1+\bar{G}_j^2)^{-1/2}$	opposite direction, after [10]

The first-order kernels integrate to zero, so G vanishes on any uniform region whatever its grey level, and it is invariant to an additive change of illumination. The finest scale is omitted because its first-order response on the 9×9 grid is dominated by pixel noise. Table 1 lists the four weightings.

Interpretation. V3 is a Schweppe-type generalised M-estimator with design function $\max(\bar{G}, 0.1)$. Its direction follows from a heteroscedastic model of the residual. Near the desired pose, $e_j \approx \boldsymbol{\ell}_j \delta\mathbf{r} + n_j$, with n_j the image noise. If the pose error is treated as random with covariance $s^2\mathbf{I}_6$, then $\operatorname{Var} e_j \approx \sigma_n^2 + s^2\|\boldsymbol{\ell}_j\|^2$, and standardising the residual calls for a divisor proportional to $(\bar{G}_j^2 + \gamma^2)^{1/2}$ with γ the ratio of noise to pose-induced variation. The floor $\max(\bar{G}, 0.1)$ is a fixed approximation of that divisor. Two consequences follow. First, the scale $\hat{s}$ of (10) is re-estimated on $\tilde{\mathbf{e}}$, so a uniform normalisation would be inert: what acts is the variation of $\bar{G}$ across the image, which reallocates the outlier budget among measurements. Second, a measurement on an untextured region has $\bar{G}$ at the floor, its residual is inflated tenfold and, if non-zero, it is rejected. V4 applies DSO's gradient-dependent design weight [10] to the Tukey weight, so that its normal equations carry $w_{\mathrm{T}}^2/(1+\bar{G}^2)$: it down-weights strongly textured measurements, the opposite of V3. An adaptive-floor variant was evaluated in the pilot and omitted from the study (Section 6.6).

6. Experimental protocol

6.1. Simulation

The scene is a planar target of 0.60×0.43 m carrying a grey-level texture. It is rendered by bilinear texture mapping (ViSP's image simulator [22]) for a free-flying camera with a focal length of 870 px, the principal point at $(160, 120)$ px and the 320×240 image. At the desired pose the camera is fronto-parallel to the target at 1.3 m. Each run applies the law (5) for 600 iterations. The desired image is always rendered from the clean texture. Occluders are painted on the texture of the current images only, so they move with the target and produce no parallax. The program builds the features and the interaction matrices itself on ViSP 3.6. With the original pixel convention it reproduces the reference implementation of [6] to every printed digit; all runs reported here use the corrected conversion from pixel to normalised coordinates.[1]

6.2. Initial poses

The 32 initial poses (seed 20261011) are drawn so that the target covers the whole image, and they are stratified into four bins of eight poses by the mean image displacement of the target between the initial and desired poses: 20–32, 32–44, 44–56 and 56–68 px (realised ranges 24–31, 34–43, 45–52 and 56–58 px). The four poses of the pilot (seed 20261012) are disjoint from them.

6.3. Scenes

Figure 1 shows the geometry. In the confirmatory campaign, an occluder of fixed geometry—the disc of [6], which covers 7.5 % of the image interior at the desired pose and 6.0–12.3 % at the initial poses—is given four appearances, so that appearance is the only variable: clean (no occluder), *dark* (grey level 0), *textured* (a patch of the same texture taken half a texture away, so that local structure is preserved and content is wrong) and *low-contrast* (the mean grey level of the texture it covers). Coverage is measured from the occluder mask by back-projecting every interior pixel onto the target plane. The *replication* uses a second texture of different content and statistics, `coffee` from scikit-image [23] (CC0; mean grey level 102 against 60, 19 % against

[1] The reference implementation passed the pixel row and column to the conversion in swapped order, which places the principal point at $(120, 160)$ in the interaction matrix. Correcting it changes the final errors by 7–21 % and no ordering.

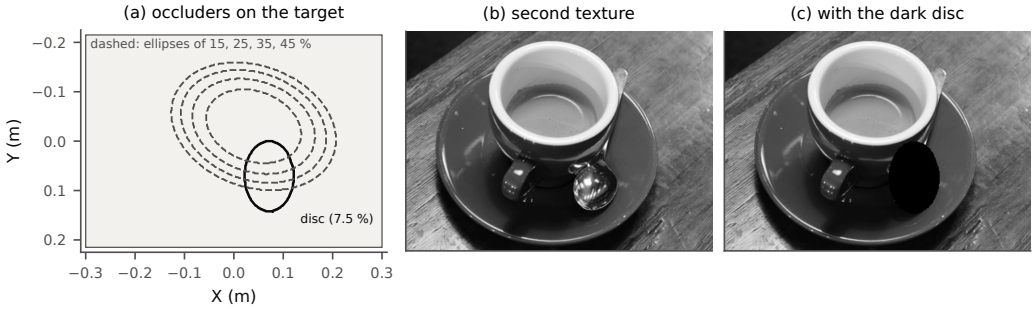


Figure 1: Scenes. (a) Occluders on the target plane: the disc of the confirmatory campaign and the replication (7.5 % of the image at the desired pose), and the four ellipses of the sweep. (b) The second texture, clean, and (c) with the dark disc. [TODO: add the reference texture of [6] as a panel, subject to its licence.]

27 % of pixels with a gradient below one grey level per pixel), with the same disc, dark, transferred through target coordinates so that its image coverage is identical. The *sweep* uses a dark ellipse on the reference texture whose size gives a coverage f of 15, 25, 35 or 45 % at the desired pose (17–73 % at the initial poses).

6.4. Outcomes

The pose error is that of the camera, the relative pose ${}^{c^*}\mathbf{M}_c$. The final camera-centre error (mm) and rotation error (deg) are the medians over the last 50 iterations.[2] A run is a *success* if both errors are below 20 mm and 1°. These thresholds were fixed after the pilot, where they separated converged runs (below 17 mm and 0.71°) from basin failures (above 440 mm and 19°). A *basin failure* is a run ending with a rotation error above 5°; it distinguishes a wrong basin from the occlusion bias of the unweighted scheme. Each run also logs, at every iteration, the rank margin $\lambda_{\min}(\mathbf{H_W})/(\mathrm{tr}(\mathbf{H_W})/6)$, the fraction of weights below 0.25 and the time spent computing the weights.

[2]The translation of the target frame seen from the camera, $\mathbf{t}({}^{c}\mathbf{M}_o) - \mathbf{t}({}^{c^*}\mathbf{M}_o)$, barely changes when the camera rotates about the target and understates the camera-centre error by about $Z\theta$: for a run that ends 175 mm from the desired camera position it reports 17 mm.

Table 2: Design. All runs: 32 initial poses, 600 iterations, both interaction matrices.

Campaign	Weightings	Scenes	Runs
Confirmatory	V1–V4	reference texture: clean; disc dark, textured, low-contrast	1 024
Replication	V2, V3	second texture: clean; disc dark	256
Sweep	V2, V3	reference texture: dark ellipse, $f = 15, 25, 35, 45\,\%$	512

6.5. Statistical analysis

The unit of analysis is the initial pose, and every comparison between two weightings is paired over poses. The effect of A against B is the Hodges–Lehmann estimate [24] of the median of the per-pose log-ratios, back-transformed to a ratio A/B, with its exact distribution-free 95 % confidence interval from the Walsh averages. The test is the two-sided Wilcoxon signed-rank test [25], exact for $n \leq 50$. Accuracy comparisons of the pre-registered families are made over the poses where both runs succeed, so that a basin failure does not enter an accuracy comparison. A test requires at least six such poses. Success is compared with the exact McNemar test [26] on paired indicators, and success rates are reported with Clopper–Pearson intervals [27]. Equivalence within $\pm 25\,\%$ is declared when the 95 % CI of the ratio lies in $[0.8, 1.25]$, which is a two-one-sided test [28] at 5 % per outcome. Multiplicity is controlled with Holm's procedure [29] within each pre-registered family. Secondary comparisons are reported with their exact 95 % CIs. These intervals are not adjusted for multiplicity and carry no confirmatory weight.

6.6. Pre-registration and design

A pilot of 160 runs on the four development poses fixed the success thresholds and the hypotheses. It also led to dropping the adaptive-floor weighting, which showed no consistent effect. The confirmatory pre-registration was then committed to the project repository before the first confirmatory run; an addendum for the replication and the sweep was committed before either was run. Table 2 summarises the three campaigns.

The confirmatory primary family comprises six tests, Holm-corrected together:

- **P1a, P1b**: V3 against V2, desired-image matrix, dark disc, camera-centre and rotation error. Prediction: V3 more accurate.

Table 3: Pre-registered primary family (Holm over six tests). Ratios are Hodges–Lehmann estimates of V3/V2 with exact 95 % CIs, over the poses where both succeed.

Test	Comparison	Result	p (Holm)	Verdict
P1a	camera-centre error, $\mathbf{L}(I^*)$, dark disc	0.658 [0.600, 0.732], $n = 29$	4.2×10^{-7}	supported
P1b	rotation error, same	0.657 [0.600, 0.732], $n = 29$	4.2×10^{-7}	supported
P2a	success, $\mathbf{L}(I)$, clean	V3-only 2, V2-only 5	1	not supported
P2b	success, $\mathbf{L}(I)$, dark disc	V3-only 2, V2-only 3	1	not supported
P2c	success, $\mathbf{L}(I)$, textured disc	V3-only 2, V2-only 3	1	not supported
P2d	success, $\mathbf{L}(I)$, low-contrast disc	V3-only 1, V2-only 6	0.50	not supported

- **P2a–d**: success of V3 against V2 with the current-image matrix, in each of the four scenes. Prediction, from the pilot: V3 succeeds less often.

The decision rule was fixed with them. If P1a and P1b are both significant with a median ratio of at most 0.8, the study is reported as a method paper on V3; otherwise as an analysis paper. The replication family (R1a, R1b) repeats P1a and P1b on the second texture, with the same criterion. The sweep family (S, 24 tests) comprises V3 against V2 on both errors (16 tests) and on success (8 tests), in each combination of fraction and matrix.

6.7. Reproducibility

The servoing program, the pose and occluder generators, the run scripts, the analysis program with its end-to-end self-test on synthetic data, the per-iteration logs and the pre-registration documents with their commit history are available at [TODO: repository URL and archive DOI]. The reference texture cannot be redistributed; the second texture is in the public domain.

7. Results

7.1. Primary family

Table 3 gives the six pre-registered tests. Both accuracy tests are significant after Holm with a median ratio of 0.66, below the 0.8 threshold, so the pre-registered decision is a method paper. None of the four success tests is significant: with the current-image matrix and the 7.5 % disc, V3 and V2 converge from similar numbers of poses (26–27 against 28–31 of 32).

Table 4: Confirmatory campaign: median final camera-centre error (mm) and rotation error (deg) of the successful runs, and number of successes out of 32. For V1 under the dark and textured discs no run succeeds; the median over all runs is given in italics.

$\widehat{\mathbf{L}}$	Scene	V1	V2	V3	V4
$\mathbf{L}(I)$	clean	5.36, 0.238 (30)	12.4, 0.541 (29)	7.99, 0.351 (26)	10.3, 0.451 (26)
	dark disc	*35.5, 1.53* (0)	0.447, 0.020 (28)	**0.327, 0.014** (27)	0.387, 0.017 (23)
	textured disc	*34.8, 1.51* (0)	0.450, 0.020 (28)	**0.329, 0.014** (27)	0.382, 0.017 (25)
	low-contrast disc	15.7, 0.678 (32)	0.985, 0.043 (31)	**0.331, 0.015** (26)	0.755, 0.033 (26)
$\mathbf{L}(I^*)$	clean	5.22, 0.230 (31)	13.4, 0.590 (31)	8.81, 0.389 (29)	12.0, 0.527 (32)
	dark disc	*31.6, 1.37* (0)	0.489, 0.022 (32)	**0.321, 0.014** (29)	0.443, 0.019 (32)
	textured disc	*31.3, 1.35* (0)	0.493, 0.022 (32)	**0.311, 0.014** (29)	0.447, 0.020 (32)
	low-contrast disc	15.0, 0.649 (31)	0.939, 0.041 (32)	**0.351, 0.015** (29)	0.894, 0.039 (32)

7.2. Accuracy

Table 4 gives the final errors and success counts of the confirmatory campaign, and Fig. 2 the V3/V2 ratio in every cell of the study where it can be computed.

Robust weighting removes the occlusion bias. Without weighting, the camera converges to a pose 31–35 mm and 1.35–1.53° from the target under the dark and textured discs. With V2 the error falls by a factor of about 65 (V2/V1 over all poses: 0.015, 95 % CI 0.012–0.020). The low-contrast disc biases V1 less (15.0–15.7 mm), within the success threshold, and V2 reduces that bias by a factor of 16. On the clean scene robustness has a cost: V2 is 2.4–2.5 times less accurate than V1 (CI 2.00–2.78), because the estimator rejects valid textured measurements while the pose error is large.

V3 is more accurate than V2 at iteration 600 in every cell. Where both converge, V3 reduces V2's final error in all eight confirmatory cells (Fig. 2). The reduction is largest under the low-contrast disc (ratios 0.40–0.42), about a third under the dark and textured discs (0.62–0.66) and a quarter to a third on the clean scene (0.69–0.74). It holds with both interaction matrices. On the clean scene V3 remains 1.5–1.7 times less accurate than V1. The structure-penalised weighting V4 lies between: its ratio to V2 is 0.89–0.94, equivalent within ±25 % in six of the eight cells.

The gain at iteration 600 grows with the occluded fraction. In the sweep, V3/V2 where both succeed is 0.77 at 15 % occlusion with either matrix (95 % CIs 0.70–0.83 and 0.72–0.80), 0.46 at 25 % (0.36–0.62 and 0.23–0.69) and

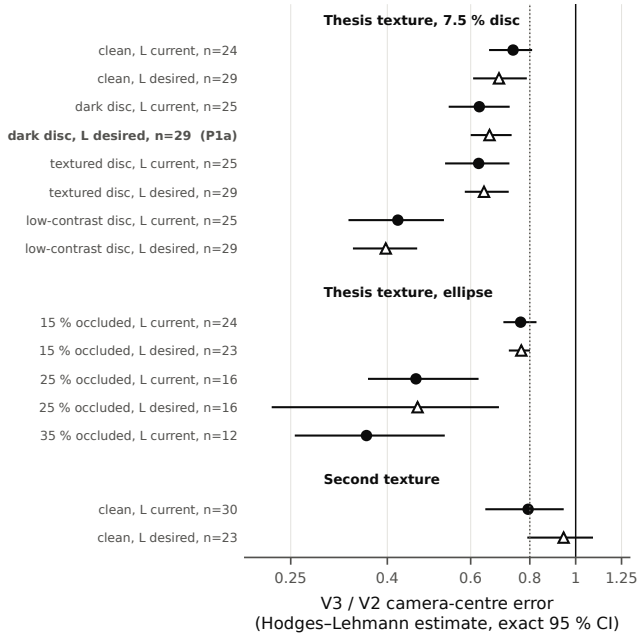


Figure 2: V3/V2 final camera-centre error over the poses where both succeed, in every cell where at least six such poses exist. Filled circles: current-image matrix; open triangles: desired-image matrix. The bold row is the pre-registered test P1a. The dotted line marks the 0.8 threshold of the decision rule. Missing cells: the occluded second texture (V2 never succeeds), the sweep at 45 % (no successes for V3 with $\mathbf{L}(I)$ or for V2 with $\mathbf{L}(I^*)$) and at 35 % with $\mathbf{L}(I^*)$ (five pairs). The rotation error gives the same ratios to two decimals.

0.36 at 35 % with the current-image matrix (0.25–0.53), all significant after Holm within family S ($p \leq 0.003$). Three cells of the family cannot be tested (Fig. 2), so its pre-specified statement that the gain holds at every fraction is not met. On the clean second texture the gain is smaller and not significant: 0.79 (0.64–0.94) with the current-image matrix and 0.94 (0.79–1.09) with the desired-image matrix.

The advantage depends on the iteration budget (exploratory). The pre-registered outcome is the error at a fixed budget of 600 iterations (24 s of simulated time). Figure 3 shows the median per-pose ratio V3/V2 at earlier budgets. Under the dark and textured discs it rises from 0.09–0.19 at iteration 100 to 0.39–0.44 at 300 and 0.62–0.64 at 600, and it is still rising at the end: between iterations 300 and 600, V2 halves its error (median factor 0.52–0.57) while V3 reduces it by only 12–15 %. The same holds in every occluded cell of the

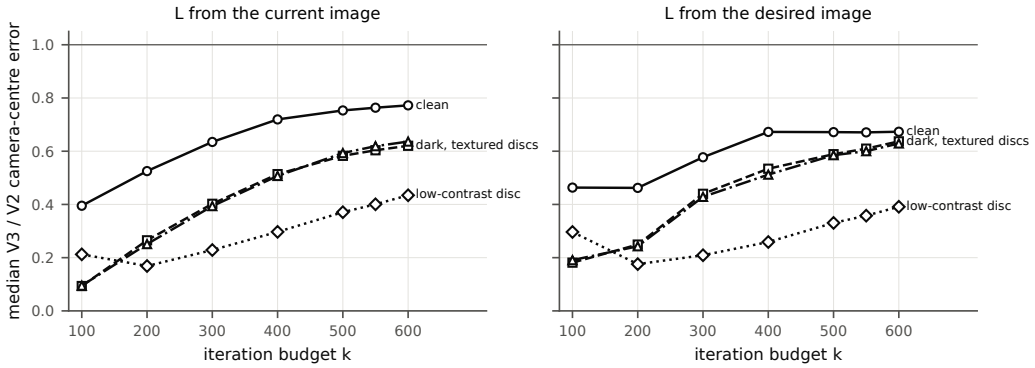


Figure 3: Median over the poses where both succeed of the per-pose ratio of V3's to V2's camera-centre error, at iteration k (each error the median over the 25 iterations up to k); confirmatory campaign. The dark and textured discs give nearly identical curves.

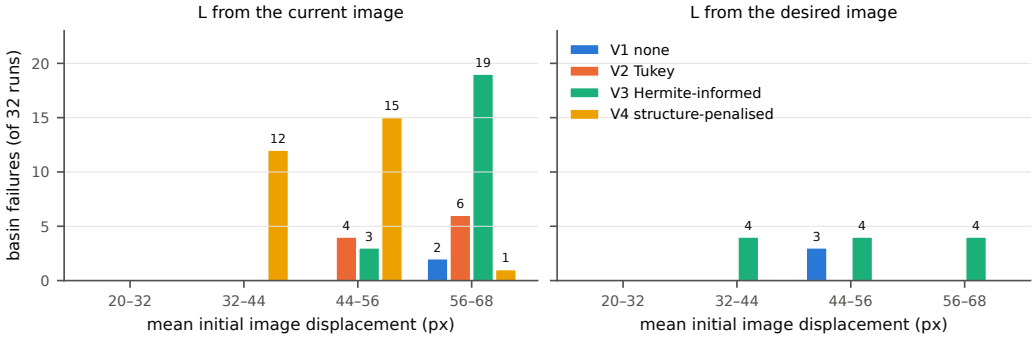


Figure 4: Basin failures (final rotation error above 5°) in the confirmatory campaign, by bin of initial image displacement; 32 runs per bin and weighting (eight poses, four scenes).

sweep (at 25 % with the current-image matrix, 0.15 at iteration 300 and 0.52 at 600). On the clean scene the ratio levels off at 0.67 (desired-image matrix) and 0.74–0.77 (current-image matrix). V3 therefore reaches a given accuracy sooner, and the larger gains at larger occluded fractions reflect at least in part the slower convergence of V2. Whether V3's error at convergence is also smaller cannot be decided from runs of this length.

7.3. Convergence

Figure 4 shows where the basin failures occur. With the current-image matrix, V3 fails from 19 of the 32 runs of the largest displacement bin and from none below 44 px; V4 fails at intermediate displacements (12 and 15

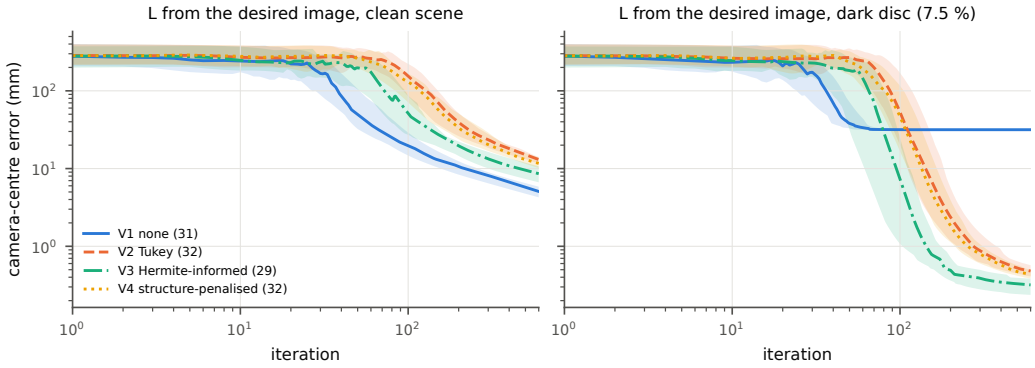


Figure 5: Median (line) and interquartile range (band) of the camera-centre error over the successful runs, desired-image matrix; the number of runs is given in the legend. On the clean scene the error decays algebraically. Under the disc V1 reaches a stationary biased pose, while the weighted schemes are still converging at iteration 600.

of 32 in bins 1 and 2); V2 fails from 10 of the 64 runs in the two largest bins. With the desired-image matrix, V2 and V4 never fail and V3 fails from three poses (poses 13, 24 and 27, in every scene). The current-image matrix thus narrows the convergence domain of the structure-based weightings, each at its own range of displacements, while the desired-image matrix, despite being exact only at the desired pose, lets V2 and V4 converge from every pose tested, at displacements up to 58 px.

Algebraic decay on the clean scene. Figure 5 shows the two regimes of Proposition 3. On the clean scene the error decays algebraically: the median slope of $\log\|\Delta\mathbf{t}\|$ against $\log k$ over $k \geq 200$ is -0.63 to -0.66 for V1 and V3 and -0.73 to -0.93 for V2 and V4. That is steeper than the $-1/2$ of the idealised analysis, but it is not geometric, and the error is still decreasing at the last iteration. Under the dark and textured discs the weighted runs end within 0.3–0.5 mm of the target, more than an order of magnitude closer than on the clean scene, because the occlusion residual keeps the gain finite. V1 reaches a stationary biased pose, as Proposition 3 predicts for fixed weights. The weighted runs are not stationary: their weights change as the robust scale contracts, which moves the equilibrium of the weighted law, and the error keeps decreasing algebraically (median slope -1.0 to -1.3 for V2, -0.25 to -0.33 for V3). The geometric regime of Proposition 3, which assumes fixed weights, is therefore observed only for V1. In both scenes the final error is a property of the iteration budget as well as of the estimator.

Table 5: Replication on the second texture: successes out of 32, and median final errors (camera centre, rotation). For V2 under the disc the median is over all runs (no success). McNemar p values are Holm-adjusted within the replication family.

$\widehat{\mathbf{L}}$	Scene	V2	V3	V3-only / V2-only	p (Holm)
$\mathbf{L}(I)$	clean	10.0 mm, 0.440° (32)	8.57 mm, 0.377° (30)	0 / 2	0.50
	dark disc	*128.1 mm, 5.62°* (0)	6.63 mm, 0.298° (24)	24 / 0	3.6×10^{-7}
$\mathbf{L}(I^*)$	clean	9.97 mm, 0.438° (31)	8.71 mm, 0.383° (24)	1 / 8	0.078
	dark disc	*174.8 mm, 7.69°* (0)	8.36 mm, 0.363° (28)	28 / 0	3.0×10^{-8}

7.4. Replication on a second texture

Table 5 gives the replication. The pre-registered tests R1a and R1b cannot be computed. Under the dark disc, V2 succeeds from none of the 32 poses with either matrix, so no pose has both runs successful. By the pre-registered criterion the primary result is therefore *not replicated*. The reason, however, is a failure of the reference method, not of V3. The pre-registered secondary success tests show V3 converging where V2 does not: from 28 against 0 poses with the desired-image matrix ($p = 3.0 \times 10^{-8}$) and from 24 against 0 with the current-image matrix ($p = 3.6 \times 10^{-7}$).

The following analysis is exploratory. V2 reaches the same final pose from every initial pose: the interquartile range of its camera-centre error is 127.7–128.3 mm with the current-image matrix and 174.7–174.9 mm with the desired-image matrix. It is an equilibrium of the weighted law, not a basin failure. From pose 1 the trajectory passes within 42 mm of the target at iteration 144 and then moves away. The offset is $\Delta\mathbf{t} = (-19.6, 171.3, 28.5)$ mm with $\theta\mathbf{u} = (7.63, 0.88, -0.10)^\circ$: a translation along y coupled with a rotation about x, $\Delta t_y \approx Z^* \theta_x$. Along this direction the image of a fronto-parallel plane changes only at second order [30]. From the same pose, the unweighted law ends 78 mm from the target and V3 8.4 mm, so V2 ends farther away than no weighting at all, and at a higher final cost (4.62×10^{10} against 4.02×10^{10} and 3.98×10^{10}). Started at the desired pose, every weighting drifts away along the same direction, so the desired pose is not an equilibrium under this occlusion.

Figure 6 locates the cause. At V2's final pose, Tukey's estimator rejects most of the disc but keeps a curved band inside it. In that band the hidden texture, the shadowed handle of the cup, is about as dark as the occluder, so the residual falls within the estimator's acceptance region. With the desired-image matrix these measurements have the non-zero rows of the hidden

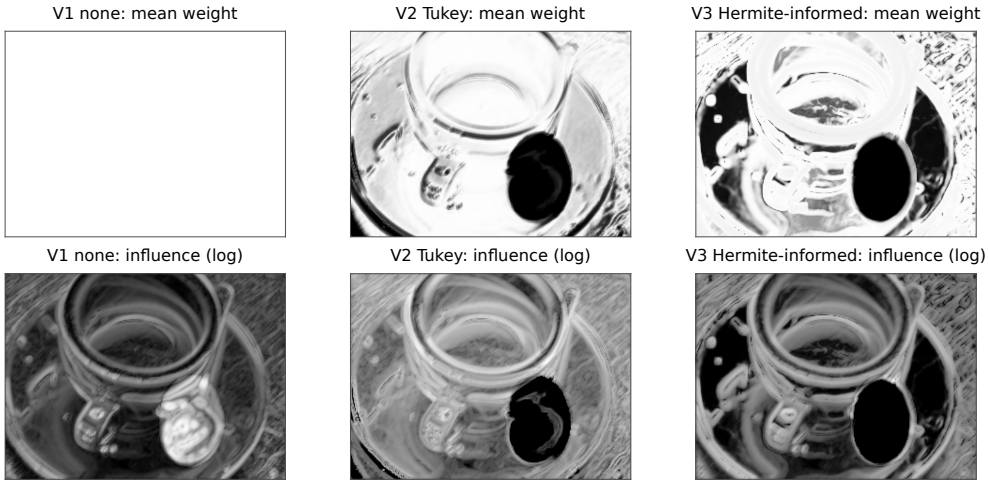


Figure 6: Second texture, dark disc, desired-image matrix, pose 1, final iteration. Top: weight averaged over the four channels at each position (white 1, black 0). Bottom: influence $\sum_k \|w_j \boldsymbol{\ell}_j\| \, |w_j e_j|$ per position, logarithmic. V2 keeps a band of measurements inside the disc, where the hidden texture is dark; V3 rejects the whole disc.

texture (Proposition 1), and they carry the influence that remains inside the occluder. V3 rejects the whole disc: $\bar{G}$ is computed from the current image, which is uniform inside the disc, so every non-zero residual there is divided by the floor and rejected. V3 thus identifies the occluder by its lack of structure, whether or not its grey level happens to match the texture it hides. With the current-image matrix the interior of the disc has zero rows, so V2's equilibrium in that case (128 mm, in the same direction) must arise from the boundary band; maps were not recorded for it. V3 is not free of bias on this texture either: it converges to 6.6–8.4 mm, against 0.33 mm on the reference texture.

7.5. Occluded fraction

Table 6 and Fig. 7 give the sweep. The two weightings break down in opposite configurations. With the desired-image matrix, V3 converges from 27–30 of 32 poses at every fraction up to 45 %, while V2 falls to 16 at 25 %, 5 at 35 % and 0 at 45 %. With the current-image matrix, V2 retains 20 of 32 at 45 % while V3 falls to 16, 14 and 0. The differences at 25, 35 and 45 % are significant after Holm in both configurations. The failures are divergences, not biased equilibria: 168 of the 198 failed runs are still moving at the last iteration, and they end 68 mm to 4.7 m and 3–175° from the target. Successful

Table 6: Sweep: successes out of 32. The 7.5 % row is the confirmatory dark disc, for reference. McNemar p values are Holm-adjusted within family S (24 tests).

	$\mathbf{L}(I)$			$\mathbf{L}(I^*)$		
f	V2	V3	p	V2	V3	p
7.5 %	28	27	–	32	29	–
15 %	27	24	0.50	27	27	1
25 %	28	16	0.0029	16	30	8.5×10^{-4}
35 %	25	14	0.022	5	28	4.1×10^{-6}
45 %	20	**0**	2.7×10^{-5}	**0**	27	2.7×10^{-7}

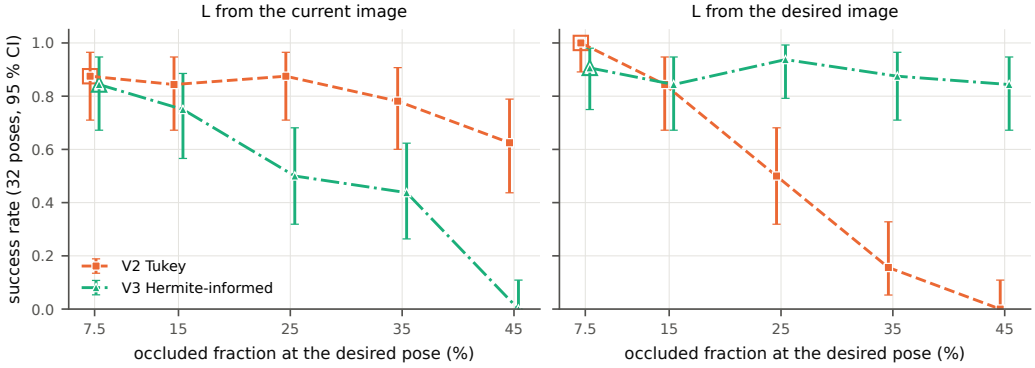


Figure 7: Success rate against the occluded fraction at the desired pose, with Clopper–Pearson 95 % intervals. The larger marker at 7.5 % is the confirmatory disc. The ordering of V2 and V3 reverses with the interaction matrix.

runs become *more* accurate at iteration 600 as the fraction grows (V3 with the desired-image matrix: 0.33, 0.23, 0.11 and 0.02 mm), consistent with a gain that the persistent residual keeps finite (Proposition 3).

The two failure modes differ; the following diagnostics are exploratory. With the desired-image matrix, V2 rejects too little. At the first iteration it gives a weight below 0.25 to only 4–10 % of the measurements, whereas the occluder covers 17–73 % of the image. At pose 1 and $f = 45$ %, its mean weight inside the occluder is 0.73, and the occluder's interior carries 55 % of the total influence on the control (V3: weight 0.36, influence 9 %). With the current-image matrix the same interior carries 4 % of the influence, as Proposition 1 requires, and V2 converges. V3 with the current-image matrix fails late. At pose 1 and $f = 45$ % it comes within 75 mm of the target at iteration 100,

then moves away along the weakly observable direction ($\Delta t_y/\theta_x = 1.29\,\mathrm{m}$), reaching 1.1 m and 51° at iteration 600, while the unweighted feature cost *decreases* (3.46×10^{11} to 3.38×10^{11} between iterations 100 and 150). Its success rate falls both with the initial coverage (0.79 below 20 %, 0.09 above 50 %) and with the displacement bin (0.59 to 0.16). The mechanism of this failure is not established.

7.6. Further measurements

Stability margin. The rank margin $\lambda_{\min}(\mathbf{H_W})/(\mathrm{tr}(\mathbf{H_W})/6)$ never fell below 5.8×10^{-6} in any of the 1 792 runs, so no failure is a loss of rank. The bound $\delta_{\mathbf{W}}$ of (7), evaluated with $\max_i(\mathbf{H_W})_{ii} \le \mathrm{tr}\,\mathbf{H_W}$, is at most 0.10 over the last 50 iterations of every successful confirmatory run and at most 0.42 in the replication, so the sufficient condition of Proposition 2 holds there. In the sweep the persistent residual keeps μ large, and the bound (median 57) is not informative.

The final cost does not rank accuracy under occlusion. Within a pose, Kendall's τ between the final feature cost $\|\mathbf{e}\|^2$ and the camera-centre error of the four weightings is 0.83–1.0 on the clean scene. Under the three occluders with the desired-image matrix it is -0.33 to -0.67, negative at 72–84 % of the poses. A lower final cost then indicates a *less* accurate pose, so the cost cannot serve to select a weighting or to declare convergence.

Cost of the weights. Per iteration, the weights take 10.5 ms for V2, 12.7 ms for V3 and 11.7 ms for V4 (medians), against 239 ms for a whole iteration of the unoptimised program. The structure measure of V3 adds 2.2 ms, under 1 % of an iteration.

8. Discussion

Two ways to fail. The results fit one account. A residual-based estimator rejects a measurement when its residual is large. An occluder's measurements do not always have large residuals: where the hidden texture is close to the occluder in grey level, they pass the test. Whether these surviving measurements matter depends on the interaction matrix. With the current-image matrix, the interior of a uniform occluder has zero rows (Proposition 1), so the retained measurements exert no force and V2 converges even at 45 % occlusion. With the desired-image matrix they have the rows of the hidden texture. They then bias the equilibrium, along the weakly observable

translation–rotation direction on the second texture, or cause divergence when the occluder is large. The structure-informed weighting removes this failure because it judges a measurement by the presence of structure in the current image, which an untextured occluder lacks whatever its grey level. Its own failure lies elsewhere: with the current-image matrix and a large occluder, the structure-informed weighting first approaches the target and then leaves it along the weakly observable direction. We have not established why. The evidence—a late departure while the unweighted cost decreases, more frequent at large displacements and large coverages—points to a weighted cost that has a spurious minimum along that direction, but it does not prove it.

Practical consequence. No weighting is robust under both interaction matrices. Within the conditions tested, V3 with the desired-image matrix is the only configuration that withstands both a large occluder (no breakdown up to 45 %) and the second texture (28 of 32 poses), and at a budget of 600 iterations it is the most accurate configuration whenever it converges. Its price is a smaller convergence domain on a clean scene: it fails from 3 of 32 poses on the reference texture and 8 of 32 on the second one, against 1 for V2 with the same matrix (not significant after Holm). With the current-image matrix, V2 is the safer choice for large occluders, at the cost of accuracy. The interaction matrix should therefore be chosen together with the weighting, not separately.

What the pre-registration changed. Three findings contradicted expectations formed before the confirmatory data. Proposition 1 suggested that, with the current-image matrix and a uniform occluder, V3 could not outperform V2, since the weights on the occluder's interior are inert. Yet V3 is more accurate in that configuration (0.63, 0.54–0.73): its gain comes through the textured measurements, which is also why it appears on the clean scene. The pilot suggested a large convergence penalty for V3 with the current-image matrix (two of four poses lost). The confirmatory campaign shows a penalty confined to the largest displacements, and P2 was not supported. The replication, designed to repeat P1, failed for an unanticipated reason that turned out to be the most informative result of the study. Finally, the pre-registered outcome, the error at a fixed budget, conflates speed and final bias. We did not anticipate this, and the exploratory analysis of Fig. 3 shows that it matters. Had the analyses been chosen after the data, the first two points would likely have been reported differently.

Limitations. The study is a simulation. The rendering is noise-free and identical for the desired and current images, the illumination is constant, and the target is a planar texture at a known constant depth, for which the interaction matrix (3) is exact at the desired pose. The occluders are painted on the target. A real occluder in front of the target would add parallax, and its texture would move differently from the target's. Two textures do not span the variety of real scenes. The second texture showed that the behaviour of V2 under occlusion depends on the texture the occluder hides, so the size of the effects reported here should not be extrapolated to other scenes. The weightings compared are Tukey-based; other influence functions, leverage-based design weights and learned weights were not studied. The gain law (6) is that of the scheme under study and shapes the clean-scene accuracy at a fixed iteration budget. The runs stop after 600 iterations, before the weighted schemes have converged under occlusion, so the comparison of final biases at convergence requires longer runs. Finally, the diagnostic analyses of failures are exploratory and rest on few runs.

9. Conclusion

In photometric visual servoing with a weighted Levenberg–Marquardt law, the interaction matrix decides where a robust weighting can act: with the current-image matrix, weights on untextured regions are inert. A structure-informed Tukey weighting, which normalises each residual by a Gauss–Hermite gradient magnitude of the current image, reduces the positioning error of Tukey weighting after 600 iterations by a third in a pre-registered comparison (ratio 0.66, $p_{\mathrm{Holm}} = 4.2 \times 10^{-7}$). The advantage is largest early in the servo and shrinks as the budget grows, so it is at least partly an advantage in speed. Paired with the desired-image interaction matrix, it is the only configuration tested that converges from most initial poses both up to 45 % occlusion and on a second texture, where Tukey weighting converges to a biased equilibrium 175 mm from the target from every pose. Residual-based rejection fails when an occluder resembles the texture it hides. Structure-based rejection does not depend on that resemblance, but with the current-image matrix it fails at large occlusion along the weakly observable direction. Longer runs, to compare the biases at convergence, validation on a robot with physical occluders, and an explanation of that last failure are the next steps.

Data availability

Code, per-run logs and analysis outputs: [TODO: repository URL and archive DOI].

Declaration of competing interest

[TODO: The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.]

Acknowledgements

[TODO: funding and acknowledgements]

Appendix A. Proofs

Proof of Proposition 1. The luminance row at q is $\boldsymbol{\ell}_I(q) = -\nabla I(q)^\top \mathbf{L}_\mathbf{x}(q)$, and by (3) $\boldsymbol{\ell}_j = \sum_q D_{k(j)}(q - p_j)\,\boldsymbol{\ell}_I(q)$, the sum running over the kernel support. If the support of the kernel combined with that of the discrete gradient operator lies in a region where I is constant, then $\nabla I(q) = \mathbf{0}$ at every q of the kernel support and $\boldsymbol{\ell}_j = \mathbf{0}$. Measurement j contributes $w_j^2 e_j \boldsymbol{\ell}_j^\top$ to $\widehat{\mathbf{L}}^\top \mathbf{W}^2 \mathbf{e}$ and $w_j^2 \boldsymbol{\ell}_j^\top \boldsymbol{\ell}_j$ to $\mathbf{H}_\mathbf{W}$. Both vanish for every w_j, and hence so does its contribution to ε and $\operatorname{diag}(\mathbf{H}_\mathbf{W})$. The velocity (5) depends on j only through the other weights, that is, through the median and the MAD of (10) and, for V3 and V4, through the mean of G. With $\widehat{\mathbf{L}} = \mathbf{L}(I^*)$ the same computation gives $\boldsymbol{\ell}_j = \sum_q D_{k(j)}(q - p_j)\,\boldsymbol{\ell}_{I^*}(q)$, which vanishes only if I^* is also uniform on the support. □

Proof of Proposition 2. Write $\mathbf{E} = \mu \operatorname{diag}(\mathbf{H}_\mathbf{W}) + \varepsilon \mathbf{I}_6 \succeq 0$, so that $\mathbf{K}_\mathbf{W} = \mathbf{H}_\mathbf{W} + \mathbf{E}$ and $\mathbf{K}_\mathbf{W}^{-1}\mathbf{H}_\mathbf{W} = \mathbf{I}_6 - \boldsymbol{\Delta}$ with $\boldsymbol{\Delta} = \mathbf{K}_\mathbf{W}^{-1}\mathbf{E}$. Since $\mathbf{K}_\mathbf{W} \succeq \mathbf{H}_\mathbf{W}$, $\|\mathbf{K}_\mathbf{W}^{-1}\| \le 1/\lambda_{\min}(\mathbf{H}_\mathbf{W})$, and $\|\boldsymbol{\Delta}\| \le \|\mathbf{K}_\mathbf{W}^{-1}\|\,\|\mathbf{E}\| \le \delta_\mathbf{W}$. On an unoccluded scene $\mathbf{e} = \mathbf{L}^*\delta\mathbf{r} + O(\|\delta\mathbf{r}\|^2)$, with $\mathbf{L}^*$ the interaction matrix at $\mathbf{r}^*$, and both choices of $\widehat{\mathbf{L}}$ satisfy $\widehat{\mathbf{L}} = \mathbf{L}^* + O(\|\delta\mathbf{r}\|)$. Hence $\widehat{\mathbf{L}}^\top\mathbf{W}^2\mathbf{e} = \mathbf{H}_\mathbf{W}\delta\mathbf{r} + O(\|\delta\mathbf{r}\|^2)$. The remainder is multiplied by $\mathbf{K}_\mathbf{W}^{-1}$, which is bounded when $\lambda_{\min}(\mathbf{H}_\mathbf{W})/\operatorname{tr}\mathbf{H}_\mathbf{W}$ is bounded away from zero. Then $\mathbf{v} = -\lambda(\mathbf{I}_6 - \boldsymbol{\Delta})\delta\mathbf{r} + O(\|\delta\mathbf{r}\|^2)$ and $\delta\mathbf{r}_{k+1} = \big((1 - T\lambda_k)\mathbf{I}_6 + T\lambda_k\boldsymbol{\Delta}_k\big)\delta\mathbf{r}_k + O(\|\delta\mathbf{r}_k\|^2)$. For $T\lambda_k \le 1$ the norm of the matrix is at most $1 - T\lambda_k + T\lambda_k\bar{\delta}$, which gives (8). No property of the sequence $\mathbf{W}_k$ other than $\delta_{\mathbf{W}_k} \le \bar{\delta}$ was used. □

Proof of Proposition 3. With $\boldsymbol{\Delta}_k \to \mathbf{0}$, the recursion of the previous proof becomes $\delta\mathbf{r}_{k+1} = (1 - T\lambda_k)\delta\mathbf{r}_k + o(\|\delta\mathbf{r}_k\|)$: the step is a scalar multiple of the error, which keeps its direction. With $\lambda_k = N_e^{0.2}\bar{e}_k = N_e^{-0.8}\|\mathbf{e}_I\|^2 = N_e^{-0.8}\rho_k(1 + o(1))$, we have $T\lambda_k = c\rho_k(1 + o(1))$ and $\rho_{k+1} = (1 - c\rho_k)^2\rho_k(1 + o(1))$. Then $1/\rho_{k+1} = (1/\rho_k)(1 + 2c\rho_k + O(\rho_k^2)) = 1/\rho_k + 2c + O(\rho_k)$, and summing gives $1/\rho_k = 2ck + o(k)$. Since $q_{\min}\|\delta\mathbf{r}\|^2 \le \rho \le q_{\max}\|\delta\mathbf{r}\|^2$ with $q_{\min}, q_{\max}$ the extreme eigenvalues of $\mathbf{Q}$, $\|\delta\mathbf{r}_k\| = \Theta(k^{-1/2})$. Under occlusion the same recursion holds about $\bar{\mathbf{r}}$ with $\lambda_k \to \lambda_\infty > 0$, and Proposition 2 gives the geometric factor. □

Appendix B. Deviations and unplanned analyses

- The study plan initially placed the adaptive-floor weighting in the primary family. The pilot showed no consistent effect, and the weighting was dropped before the confirmatory pre-registration.

- The replication tests R1a and R1b could not be computed (no pose with both runs successful). They are reported as not replicated, as the pre-registered criterion requires.

- Three cells of the sweep family have fewer than six pairs and no p value, as the addendum specifies.

- The following analyses are exploratory and were not pre-registered: basin failures by displacement bin; the seven diagnostic runs on the second texture (from the desired pose and from pose 1, with weight and influence maps); the four diagnostic runs at 45 % occlusion; the stability bound $\delta_{\mathbf{W}}$; Kendall's τ between cost and accuracy; tail slopes; the dependence of the V3/V2 ratio on the iteration budget.

References

[1] C. Collewet, E. Marchand, F. Chaumette, Visual servoing set free from image processing, in: Proceedings of the IEEE International Conference on Robotics and Automation (ICRA), 2008, pp. 81–86.

[2] C. Collewet, E. Marchand, Photometric visual servoing, IEEE Transactions on Robotics 27 (4) (2011) 828–834.

[3] A. Dame, E. Marchand, Mutual information-based visual servoing, IEEE Transactions on Robotics 27 (5) (2011) 958–969.

[4] N. Crombez, E. M. Mouaddib, G. Caron, F. Chaumette, Visual servoing with photometric Gaussian mixtures as dense features, IEEE Transactions on Robotics 35 (1) (2019) 49–63.

[5] M. Bakthavatchalam, O. Tahri, F. Chaumette, A direct dense visual servoing approach using photometric moments, IEEE Transactions on Robotics 34 (5) (2018) 1226–1239.

[6] TODO: author's prior publication on Hermite photometric visual servoing (thesis Chapter 3), Hermite photometric visual servoing, replace with the published reference, or with the thesis (TODO).

[7] A. I. Comport, E. Marchand, F. Chaumette, Statistically robust 2-D visual servoing, IEEE Transactions on Robotics 22 (2) (2006) 415–420.

[8] E. Malis, E. Marchand, Experiments with robust estimation techniques in real-time robot vision, in: Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2006, pp. 223–228.

[9] J. Engel, T. Schöps, D. Cremers, LSD-SLAM: Large-scale direct monocular SLAM, in: Computer Vision – ECCV 2014, Vol. 8690 of Lecture Notes in Computer Science, Springer, 2014, pp. 834–849.

[10] J. Engel, V. Koltun, D. Cremers, Direct sparse odometry, IEEE Transactions on Pattern Analysis and Machine Intelligence 40 (3) (2018) 611–625.

[11] E. Handschin, F. C. Schweppe, J. Kohlas, A. Fiechter, Bad data analysis for power system state estimation, IEEE Transactions on Power Apparatus and Systems PAS-94 (2) (1975) 329–337.

[12] F. R. Hampel, E. M. Ronchetti, P. J. Rousseeuw, W. A. Stahel, Robust Statistics: The Approach Based on Influence Functions, Wiley, 1986.

[13] F. Chaumette, S. Hutchinson, Visual servo control, Part I: Basic approaches, IEEE Robotics & Automation Magazine 13 (4) (2006) 82–90.

[14] E. Malis, Improving vision-based control using efficient second-order minimization techniques, in: Proceedings of the IEEE International Conference on Robotics and Automation (ICRA), 2004, pp. 1843–1848.

[15] D. W. Marquardt, An algorithm for least-squares estimation of nonlinear parameters, Journal of the Society for Industrial and Applied Mathematics 11 (2) (1963) 431–441.

[16] J.-B. Martens, The Hermite transform—theory, IEEE Transactions on Acoustics, Speech, and Signal Processing 38 (9) (1990) 1595–1606.

[17] S. Baker, I. Matthews, Lucas-Kanade 20 years on: A unifying framework, International Journal of Computer Vision 56 (3) (2004) 221–255.

[18] R. A. Maronna, R. D. Martin, V. J. Yohai, M. Salibián-Barrera, Robust Statistics: Theory and Methods (with R), 2nd Edition, Wiley, 2019.

[19] B. A. Nosek, C. R. Ebersole, A. C. DeHaven, D. T. Mellor, The preregistration revolution, Proceedings of the National Academy of Sciences 115 (11) (2018) 2600–2606.

[20] A. E. Beaton, J. W. Tukey, The fitting of power series, meaning polynomials, illustrated on band-spectroscopic data, Technometrics 16 (2) (1974) 147–185.

[21] P. J. Rousseeuw, C. Croux, Alternatives to the median absolute deviation, Journal of the American Statistical Association 88 (424) (1993) 1273–1283.

[22] E. Marchand, F. Spindler, F. Chaumette, ViSP for visual servoing: a generic software platform with a wide class of robot control skills, IEEE Robotics & Automation Magazine 12 (4) (2005) 40–52.

[23] S. van der Walt, J. L. Schönberger, J. Nunez-Iglesias, F. Boulogne, J. D. Warner, N. Yager, E. Gouillart, T. Yu, scikit-image: image processing in Python, PeerJ 2 (2014) e453.

[24] J. L. Hodges, E. L. Lehmann, Estimates of location based on rank tests, The Annals of Mathematical Statistics 34 (2) (1963) 598–611.

[25] F. Wilcoxon, Individual comparisons by ranking methods, Biometrics Bulletin 1 (6) (1945) 80–83.

[26] Q. McNemar, Note on the sampling error of the difference between correlated proportions or percentages, Psychometrika 12 (2) (1947) 153–157.

[27] C. J. Clopper, E. S. Pearson, The use of confidence or fiducial limits illustrated in the case of the binomial, Biometrika 26 (4) (1934) 404–413.

[28] D. J. Schuirmann, A comparison of the two one-sided tests procedure and the power approach for assessing the equivalence of average bioavailability, Journal of Pharmacokinetics and Biopharmaceutics 15 (6) (1987) 657–680.

[29] S. Holm, A simple sequentially rejective multiple test procedure, Scandinavian Journal of Statistics 6 (2) (1979) 65–70.

[30] F. Chaumette, Potential problems of stability and convergence in image-based and position-based visual servoing, in: The Confluence of Vision and Control, Vol. 237 of Lecture Notes in Control and Information Sciences, Springer, 1998, pp. 66–78.